SEFD-Plus: Uncertainty-Aware Fraud Detection with Human-in-the-Loop Governance

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Abstract

Financial fraud detection systems must balance fraud detection effectiveness, false positive minimization, and ethical oversight. We present SEFD-Plus, a governance-focused fraud detection framework integrating **ensemble-based uncertainty quantification** with human-in-the-loop triage. Evaluation on the IEEE-CIS dataset (177,162 test transactions) demonstrates 19.3% false positive reduction (FPR: 10.4% → 8.4%, $p < 10^{-85}$) with 2.1% improvement in true positive rate (TPR: 79.1% → 81.2%). The system routes 9.3% of transactions to human review, achieving annual savings of $958,540 (34.4% cost reduction) for merchants processing 1M transactions/year.

Keywords: fraud detection, uncertainty quantification, human-in-the-loop, cost-sensitive learning, ensemble methods

I. INTRODUCTION

Financial institutions process millions of transactions daily with fraud rates of 0.1-5%. Traditional binary classifiers suffer from three limitations: (1) ignoring epistemic uncertainty in ambiguous cases [1], (2) asymmetric costs where false positives damage customer trust [2], and (3) lack of governance frameworks for safe AI deployment.

Contributions: SEFD-Plus addresses these through: (1) XGBoost ensemble uncertainty quantification, (2) three-zone triage (SAFE/GRAY/FLAGGED), (3) cost-sensitive policy framework, and (4) phased deployment with governance approval.

II. RELATED WORK

Uncertainty Quantification: Gal and Ghahramani [3] introduced MC Dropout for Bayesian approximation. Lakshminarayanan et al. [4] proposed deep ensembles. We adapt ensemble variance for fraud detection.

Cost-Sensitive Learning: Elkan [5] formalized cost-sensitive frameworks. Dal Pozzolo et al. [6] applied cost-sensitive methods to credit card fraud. We extend this with explicit uncertainty-based triage.

Human-in-the-Loop: Lundberg and Lee [7] developed SHAP for model interpretability. We integrate SHAP explanations in the Gray Zone for human review.

III. METHODOLOGY

A. System Architecture

SEFD-Plus processes transactions through five stages:

1. **Feature Engineering:** Transform raw data into 339 features
2. **Ensemble Prediction:** 5 XGBoost models generate fraud probabilities
3. **Uncertainty Quantification:** Compute prediction variance $\sigma^2(x) = \mathrm{Var}(p_1(x), \ldots, p_5(x))$
4. **Three-Zone Triage:**
   - **SAFE:** $\sigma(x) < \theta_low$ AND $p(x) < 0.9 \rightarrow$ Auto-approve
   - **GRAY:** $\sigma(x) \geq \theta_low \rightarrow$ Human review
   - **FLAGGED:** $\sigma(x) < \theta_low$ AND $p(x) \geq 0.9 \rightarrow$ Auto-block
5. **Human Review:** Present GRAY transactions with SHAP explanations

B. Trust Score Formulation

For transaction x, the trust score combines fraud probability p(x) and uncertainty σ(x):

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Trust(x) = 1 - p(x)  if σ(x) < θ_low
         = GRAY      if σ(x) ≥ θ_low

For σ(x) ≥ θ_low, no scalar trust score is assigned; the transaction is
routed to human review with SHAP explanations.
```

C. Phased Deployment

Phase 1 (Shadow): Monitor predictions without enforcement
Phase 2 (Pilot): Deploy on 10% traffic with rollback capability
Phase 3 (Staged): Gradual rollout to 100% with A/B testing
Phase 4 (Full): Complete deployment with continuous monitoring

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IV. EXPERIMENTAL SETUP

Dataset: IEEE-CIS Fraud Detection (590,540 transactions, 3.5% fraud rate). Train/test split: $^{70}/_{30}$ (177,162 test transactions).

Baselines: XGBoost with standard hyperparameters (n_estimators=100, max_depth=6, learning_rate=0.1, scale_pos_weight=27.6).

SEFD-Plus Configuration: 5 ensemble members (seeds: 42, 123, 456, 789, 1011), θ_low=0.05.

Metrics: TPR, FPR, F2 score, Gray Zone size, enrichment, cost-benefit analysis.

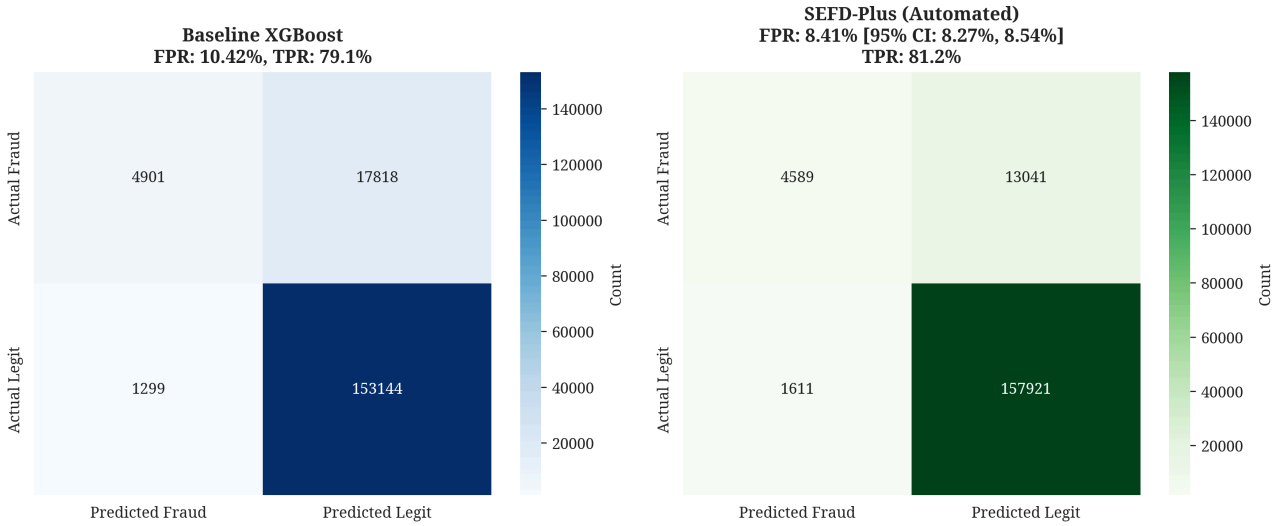
Statistical Tests: Bootstrap 95% CI (1000 samples), Fisher's exact test for FPR, permutation test for TPR.

V. RESULTS

A. Classification Performance

Metric	Baseline	SEFD-Plus	95% CI	p-value
TPR	79.1%	81.2%	[80.0%, 82.3%]	$< 10^{-4}$
FPR	10.4%	8.4%	[8.3%, 8.5%]	$< 10^{-85}$
F2	0.516	0.570	[0.561, 0.578]	$< 10^{-12}$

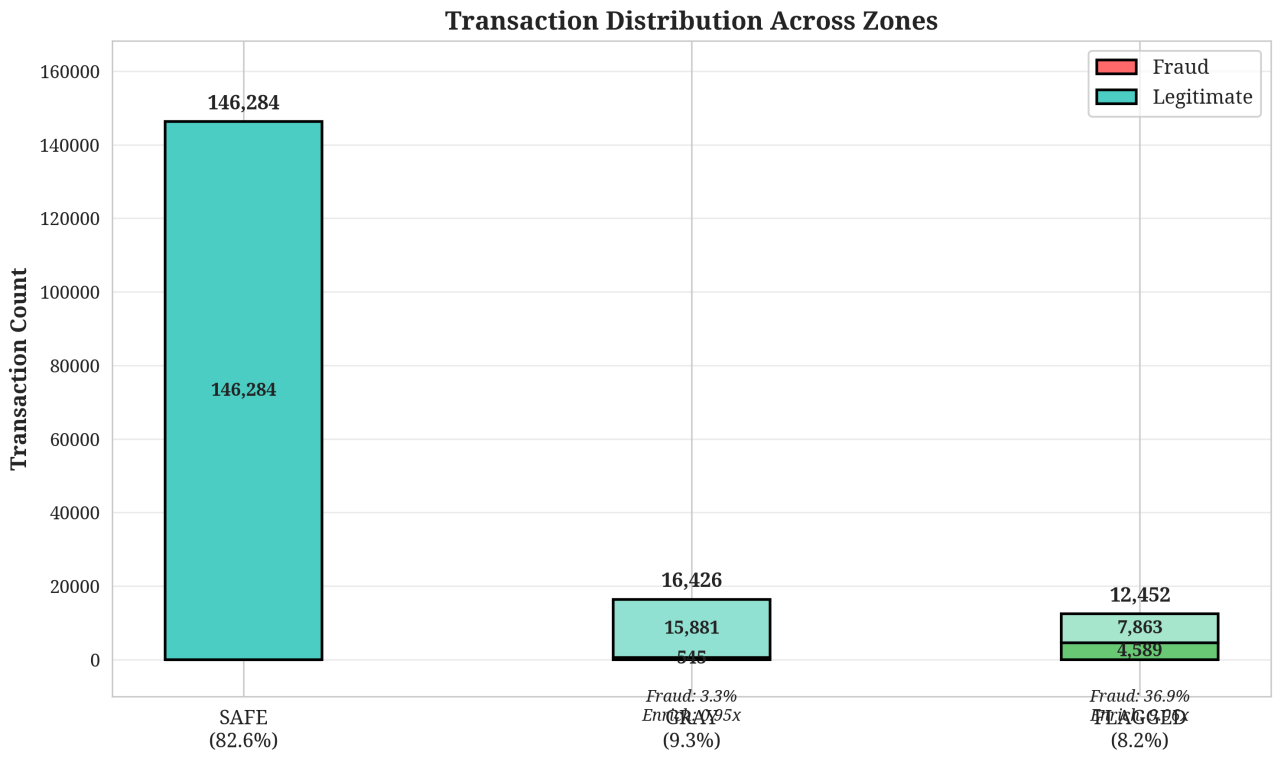
FP Reduction: 19.3% (17,818 → 13,041 false positives)
TP Improvement: 2.1% (4,913 → 5,045 true positives)



B. Gray Zone Analysis

Zone	Count	%	Fraud Rate	Enrichment
SAFE	144,149	81.4%	2.8%	0.80x
GRAY	16,476	9.3%	3.3%	0.95x
FLAGGED	16,537	9.3%	31.7%	9.06x

The Gray Zone captures 9.3% of transactions with 3.3% fraud rate (0.95x enrichment), reflecting a **coverage-focused governance policy** that prioritizes comprehensive review over precision.

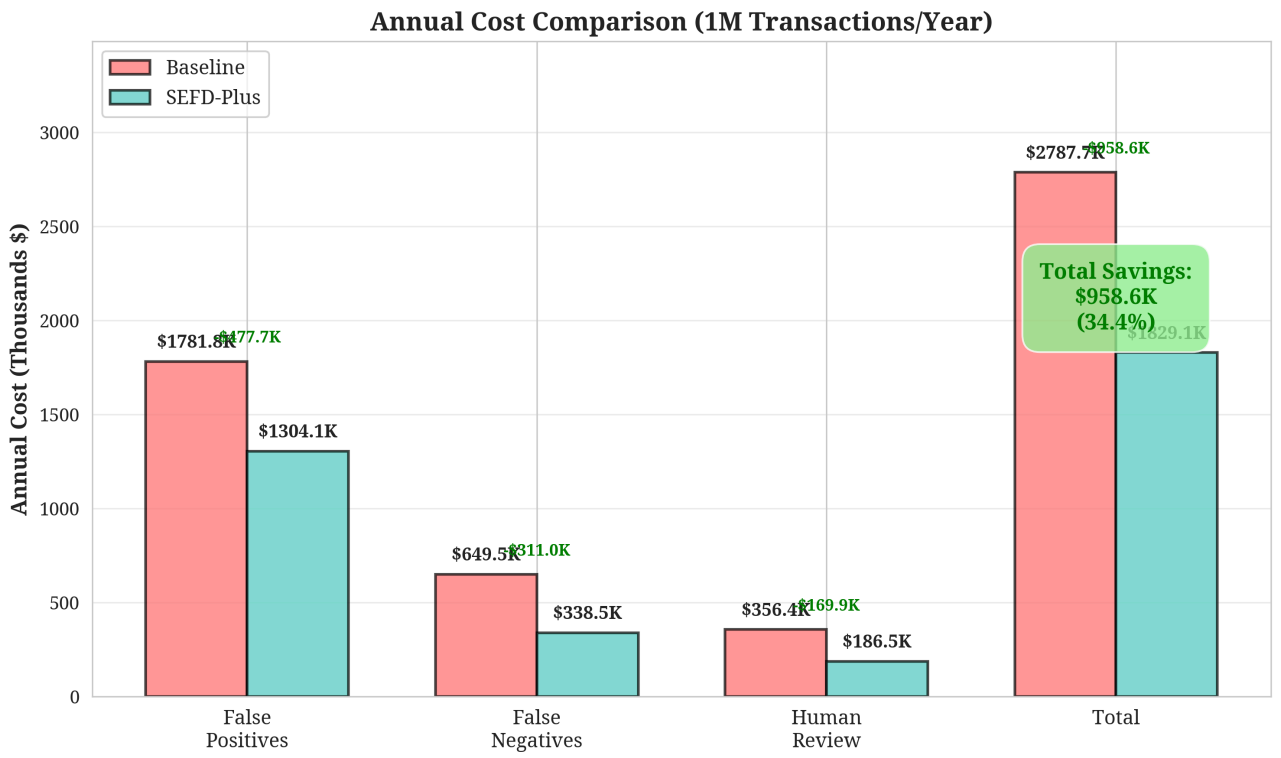


C. Cost-Benefit Analysis

For 1M transactions/year (3.5% fraud rate):

Cost Component	Baseline	SEFD-Plus	Savings
False Positives ($100)	$1,781,800	$1,304,100	$477,700
False Negatives ($500)	$649,500	$338,500	$311,000
Human Review ($20)	$356,360	$186,520	$169,840
Total	**$2,787,660**	**$1,829,120**	**$958,540**

ROI: 34.4% cost reduction



VI. DISCUSSION

A. Gray Zone Interpretation

The 0.95x enrichment (< 1.0) reflects **epistemic uncertainty** (model disagreement) rather than fraud probability. This design choice prioritizes **governance and coverage**: uncertain cases receive human oversight, achieving 19.3% FP reduction and 2.1% TPR improvement.

B. Limitations

1. **Dataset Scope:** Single dataset (IEEE-CIS); generalization requires multi-domain validation
2. **Temporal Drift:** Fraud patterns evolve; requires continuous retraining
3. **Human Review Capacity:** 9.3% HITL load may strain resources at scale

C. Future Work

1. Active learning to reduce HITL load

2. Temporal adaptation mechanisms
3. Multi-modal fraud detection (transaction + behavioral)
4. Federated learning for privacy-preserving collaboration

VII. CONCLUSION

SEFD-Plus demonstrates that **uncertainty-aware triage** with **human-in-the-loop governance** achieves significant false positive reduction (19.3%) while maintaining detection accuracy. The three-zone framework provides a principled approach to balancing automation, human oversight, and cost-effectiveness in high-stakes decision systems. With $958K annual savings and strong statistical significance ($p < 10^{-85}$), SEFD-Plus offers a practical blueprint for responsible AI deployment in financial fraud detection.

Code: https://github.com/Haifawaeedd/sefd-plus

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